Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/15/2019
PI Name:	Wang, Kegang Ph.D.		
Project Title:	Testing Analytical and Numerical Models in Phase Coarsening using NASA Physical Sciences Informatics System		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
Start Date:	09/16/2016	**End Date:**	09/15/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/15/2020 per NSSC information (Ed., 6/18/20) NOTE: End date changed to 9/15/2019 per NSSC information (Ed., 6/17/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Glicksman, Martin Ph.D. (Florida Institute of Technology)		
Grant/Contract No.:	NNX16AT74G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A quantitative understanding of microstructure evolution by phase coarsening is crucial to the optimization of processing procedures for controlling the final structure and properties of multiphase materials. The volume fraction of the coarsening phase plays an integral role in the kinetics. The existing theoretical work has been limited to some volume fractions. The most ground-based experimental study of phase coarsening has been limited in model alloy systems. There were four microgravity experimental works for phase coarsening in Sn-Pb mixtures. The microgravity experiments greatly reduced the rate of sedimentation and allowed the dynamics of the phase coarsening to be investigated carefully and accurately. The specific thrusts of this project therefore lie in advancing the state of the art of understanding phase coarsening using NASA Physical Sciences Informatics (PSI) System. The proposed research project is to test existing analytical and numerical models for phase coarsening using data in NASA Physical Sciences Informatics System. The primary goals of this project are: (1) We will test the diffusion screening theory of phase coarsening by virtue of the microgravity experiments in NASA Physical Sciences Informatics System; (2) We will test numerical modeling such as phase-field simulation; (3) Large-scale three-dimensional phase-field simulations will be conducted utilizing the microgravity experiments in NASA Physical Sciences Informatics System; (4) We will carry out a three-way comparison among theory, simulation, and microgravity experiments; (5) After the testing, an improved theory and a more accurate numerical modeling may be developed if needed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research is potentially improving our understanding of fundamental materials phenomena, and advancing the state of the art in both theory and experiment as it pertains to microstructural evolution in materials science. Our research will help engineers improve process designs in economically important areas, including liquid-phase sintering for the production of hard tool materials, and industrial crystallization of commodity chemicals and fertilizers, where controlling the particle size distribution underlies important safety and manufacturing issues. The improved knowledge of phase coarsening in this project will have significant impact on alloy design and service-life strategies for high-temperature materials. Our work has affected multiple disciplines such as materials science, chemistry, physics, and nanotechnology within the field of statistical mechanics modeling for pattern formation. This simulation is of significant interest for the physical sciences and technologies community.
Task Progress:	The Principal Investigator (PI) Wang and Co-PI Glicksman at Florida Institute of Technology have worked on this project during this report period. The specific objective of this research is to test existing theories for phase coarsening, including evaluating the authors' contribution of diffusion-screening theory, by performing numerical modeling of phase coarsening using data resident in NASA's PSI system. Specific research activities include: (1) use data archived in NASA's PSI system from the Coarsening in Solid-Liquid Mixtures (CSLM) microgravity experiments, from which we determine mean and maximum particle sizes in each sample processed through the CSLM microgravity experiments; (2) apply a two-dimensional (2-D) phase-field simulation code, to allow simulation and evolution of initial microstructures that were observed in the CSLM experiments for Sn-Pb solid-liquid phase mixtures, and which provide input data for our phase-field code to numerically evolve coarsened microstructures over time; (3) calculate particle size distributions (PSDs) from the prior NASA CSLM experiments, our phase-field and multiparticle diffusion simulations, and from diffusion screening theory; (4) determine the coarsening rate constant for these binary alloys by tracking their microstructure evolution as initially recorded from the CSLM experiments in NASA's PSI system, and from subsequent phase-field simulations. We successfully developed a 2-D phase-field computer code, and with it conducted 2-D phase-field simulations for phase coarsening at volume fractions of 0.7 and 0.3. We compared these simulation results with archived data derived originally from NASA’s CSLM experiments. In addition, we extracted solute concentration distributions from the initial microstructures measured at the earliest processing time as archived from the CSLM experiments for alloys at volume fractions of 0.7 and 0.3. To allow quantitative comparison of our phase coarsening simulations with archived experimental data from CSLM, NASA data were used as inputs for our phase-field simulations, which allowed simulated evolution of coarsened microstructures for the Sn-Pb alloy mixtures processed aboard the International Space Station (ISS). By contrast, we show that coarsened microstructures from the CSLM experiments and the computed microstructure generated with our phase-field simulation at 24 hours and at 30,000 simulation steps, at volume fraction of 0.3, respectively. We also show that microstructures from the CSLM experiments and the microstructure we computed from our phase-field simulation at 24 hours and at 30,000 simulation steps, at volume fractions of 0.7, respectively. We compared maximum particle sizes from the CSLM experiments and simulations with theoretical predictions from our diffusion screening theory. In order to check the accuracy of this advanced theory of phase coarsening, we measured values of the scaled maximum particle sizes, (ρ max=Rmax/<R>), where Rmax is the largest radius observed in the coarsened population, and <R> is the average particle radius measured for that population. For example, for the volume fraction of 0.3, simulation and diffusion screening theory predicted that ρ max=1.77, and 1.73, respectively, and CSLM experiment yielded ρ max=1.84. These values are in good agreement given both the uncertainties in the statistical experimental and theoretical estimates. We show the corresponding particle size distributions, PSDs, obtained from microgravity experiments and our phase-field simulations at volume fractions of 0.7 and 0.3, respectively. These comparisons are encouraging.
Bibliography Type:	Description: (Last Updated: 10/30/2021)
Abstracts for Journals and Proceedings	Wang KG. "Effects of Phase Coarsening on Mechanical Properties of Materials." Poster at Seminar, Beijing Institute of Technology, June 21, 2019. Seminar, Beijing Institute of Technology, June 21, 2019. , Jun-2019

Abstracts for Journals and Proceedings	Wang KG. "Advances in Ostwald ripening in materials." Poster at Seminar, Wuhan Institute of Technology, China, December 11, 2018. Seminar, Wuhan Institute of Technology, China, December 11, 2018. , Dec-2018